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JEE MAIN · QUESTION PAPER

# JEE Main 2021 (Online) 26th August Evening Shift

JEE Main Question Paper with Answer Key

| 26 Aug 2021 | 90 | 19 | English |
|---|---|---|---|
| HELD ON | QUESTIONS | OUT OF SYLLABUS | LANGUAGE |

INSIDE THIS PDF

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| Chemistry | 21 questions |
| Mathematics | 23 questions |
| Physics | 27 questions |
| Out of syllabus | 19 questions |
| Answer key | at the end |

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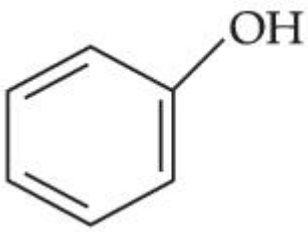
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

## Chemistry 21 questions

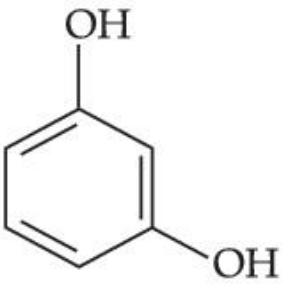
**Q1.** MCQ (Single Correct Answer)

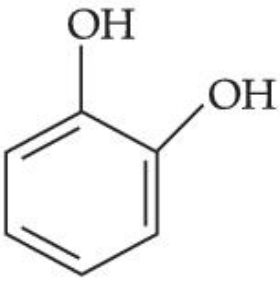
Which one of the following phenols does not give colour when condensed with phthalic anhydride in presence of conc. $H_2SO_4$ ?

Open solution

(A) 


(B) 


(C) 


(D) 


**Q2.** MCQ (Single Correct Answer)

The bond order and magnetic behaviour of $O_2^-$ ion are respectively :

Open solution

(A) 1.5 and paramagnetic

(B) 1.5 and diamagnetic

(C) 2 and diamagnetic

(D) 1 and paramagnetic

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**Q3.** MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Sucrose is a disaccharide and a non-reducing sugar.

Reason (R) : Sucrose involves glycosidic linkage between $C_1$ of $\beta$-glucose and $C_2$ of $\alpha$-fructose.

Choose the most appropriate answer from the options given below :

**(A)** Both (A) and (R) are true but (R) is not the true explanation of (A).

**(B)** (A) is false but (R) is true.

**(C)** (A) is true but (R) is false.

**(D)** Both (A) and (R) are true and (R) is the true explanation of (A)

**Q4.** MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

| | List - I (Chemical Reaction) | | List - II (Reagent used) |
|---|---|---|---|
| (a) | $CH_3COOCH_2CH_3 \rightarrow CH_3CH_2OH$ | (i) | $CH_3MgBr/H_3O^+$ (1.equivalent) |
| (b) | $CH_3COOCH_3 \rightarrow CH_3CHO$ | (ii) | $H_2SO_4/H_2O$ |
| (c) | $CH_3C \equiv N \rightarrow CH_3CHO$ | (iii) | DIBAL-H/$H_2O$ |
| (d) | $CH_3C \equiv N \rightarrow CH_3$–C(=O)–$CH_3$ | (iv) | $SnCl_2$, $HCl/H_2O$ |

Choose the most appropriate match :

**(A)** a-ii, b-iv, c-iii, d-i

**(B)** a-iv, b-ii, c-iii, d-i

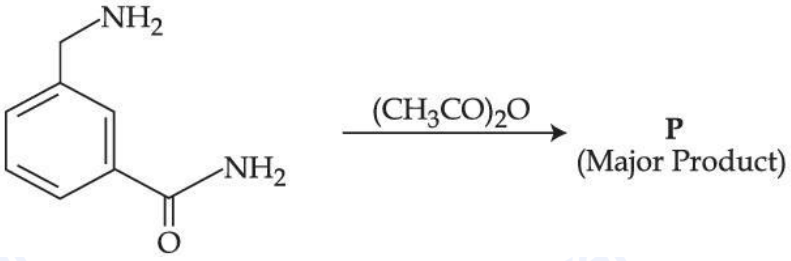
**(C)** a-ii, b-iii, c-iv, d-i

**(D)** a-iii, b-ii, c-i, d-iv

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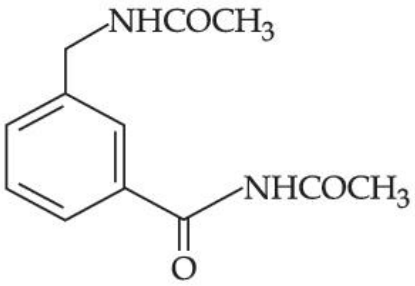
**Q5.** MCQ (Single Correct Answer)

Open solution

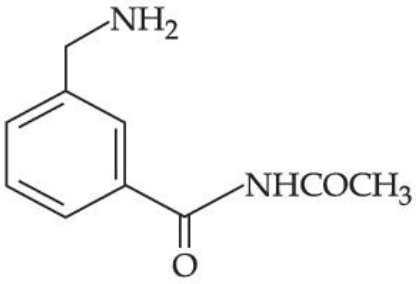



The major product in the above reaction is :

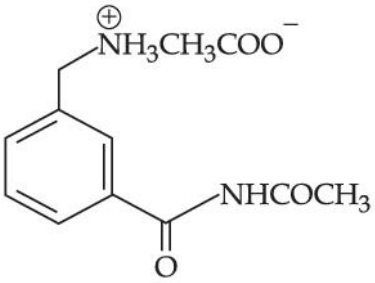
(A)



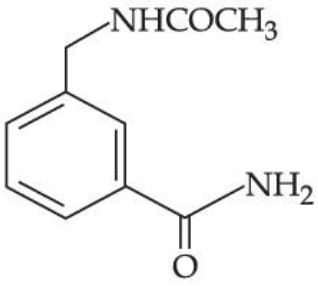

(B)




(C)




(D)




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**Q6.** MCQ (Single Correct Answer)

Indicate the complex/complex ion which did not show any geometrical isomerism :

(A) $[CoCl_2(en)_2]$ (B) $[Co(CN)_5(NC)]^{3-}$ (C) $[Co(NH_3)_3(NO_2)_3]$ (D) $[Co(NH_3)_4Cl_2]^+$

Open solution

**Q7.** MCQ (Single Correct Answer)

Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value.

Complexes : $\underset{A}{[CoF_6]^{3-}}, \underset{B}{[Co(H_2O)_6]^{2+}}, \underset{C}{[Co(NH_3)_6]^{3+}} and \underset{D}{[Co(en)_3]^{3+}}$

Choose the correct option :

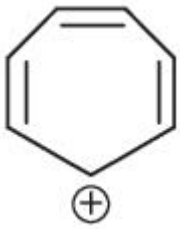
(A) A < B < C < D (B) B < A < C < D (C) B < C < D < A (D) C < D < B < A

Open solution

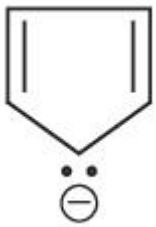
**Q8.** MCQ (Single Correct Answer)

Which one of the following compounds is not aromatic?

(A)



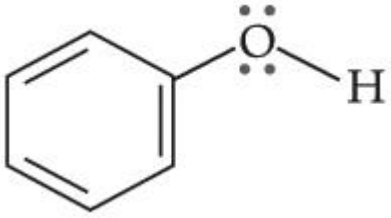
(B)



(C)



(D)




Open solution

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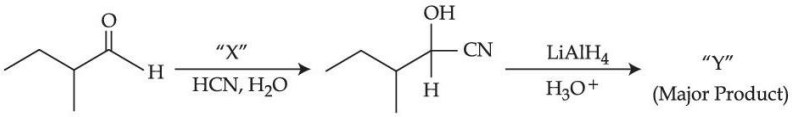
**Q9.** MCQ (Single Correct Answer)

The number of stereoisomers possible for 1, 2-dimethyl cyclopropane is :

(A) One (B) Four (C) Two (D) Three

Open solution

**Q10.** MCQ (Single Correct Answer)

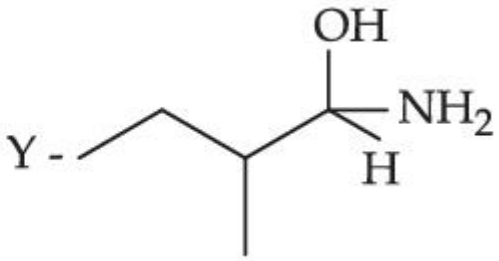



Open solution

Consider the given reaction, identify 'X' and 'Y' :

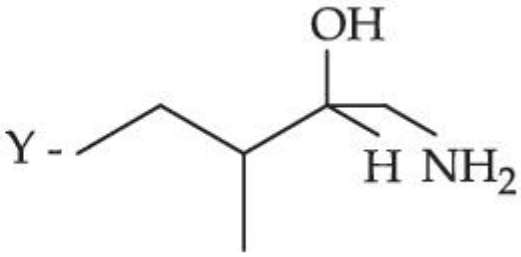
(A)

X - NaOH



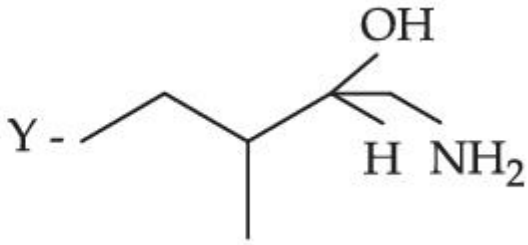

(B)

X - $HNO_3$



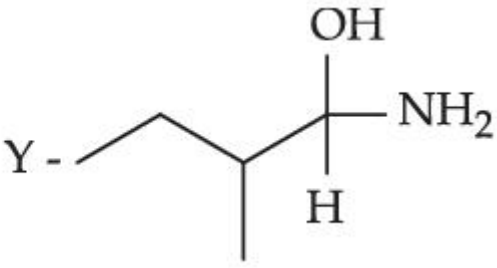

(C)

X - NaOH



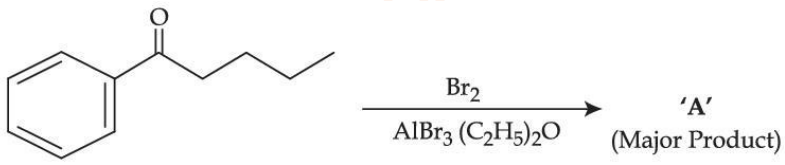

(D)

X - $HNO_3$




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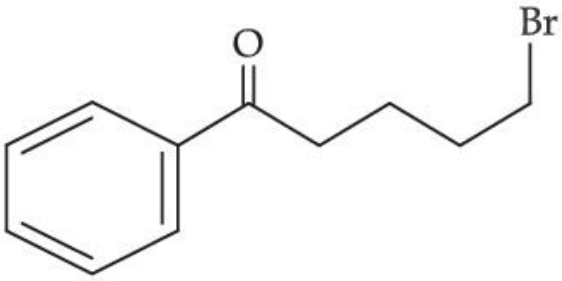
**Q11.** MCQ (Single Correct Answer)



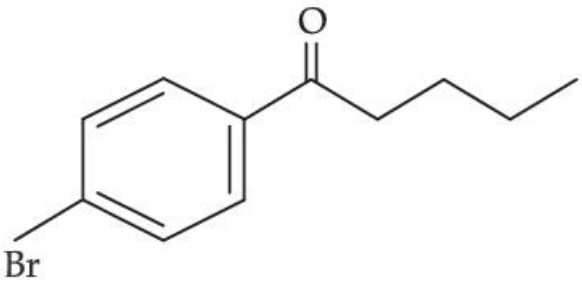

Open solution

Consider the given reaction, the product A is :

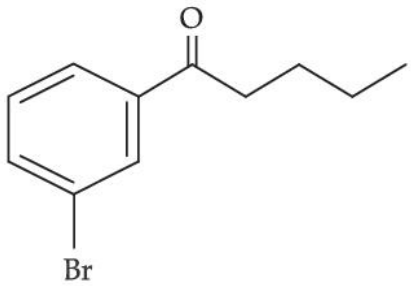
(A)



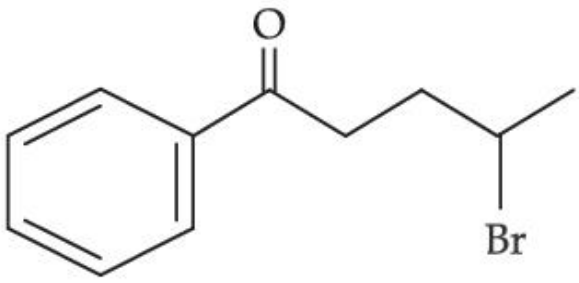

(B)




(C)




(D)




**Q12.** Numerical

In the sulphur estimation, 0.471 g of an organic compound gave 1.44 g of barium sulphate. The percentage of sulphur in the compound is _____%. (Nearest integer)

(Atomic Mass of Ba = 137 u)

Open solution

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**Q13.** Numerical

The equilibrium constant $K_c$ at 298 K for the reaction A + B $\rightleftharpoons$ C + D is 100. Starting with an equimolar solution with concentrations of A, B, C and D all equal to 1M, the equilibrium concentration of D is _____ $\times 10^{-2}$ M. (Nearest integer)

Open solution

**Q14.** Numerical

For water $\Delta_{vap}$ H = 41 kJ $mol^{-1}$ at 373 K and 1 bar pressure. Assuming that water vapour is an ideal gas that occupies a much larger volume than liquid water, the internal energy change during evaporation of water is _____ kJ $mol^{-1}$

[Use : R = 8.3 J $mol^{-1}$ $K^{-1}$]

Open solution

**Q15.** Numerical

A metal surface is exposed to 500 nm radiation. The threshold frequency of the metal for photoelectric current is 4.3 $\times 10^{14}$ Hz. The velocity of ejected electron is _____ $\times 10^{5}$ $ms^{-1}$ (Nearest integer)

[Use : h = 6.63 $\times 10^{-34}$ Js, $m_e$ = 9.0 $\times 10^{-31}$ kg]

Open solution

**Q16.** Numerical

For the galvanic cell,

Zn(s) + $Cu^{2+}$ (0.02 M) $\rightarrow$ $Zn^{2+}$ (0.04 M) + Cu(s),

$E_{cell}$ = _____ $\times 10^{-2}$ V. (Nearest integer)

[Use : $E^0_{Cu/Cu^{2+}}$ = – 0.34 V, $E^0_{Zn/Zn^{2+}}$ = + 0.76 V, $\frac{2.303RT}{F} = 0.059\,V$]

Open solution

**Q17.** Numerical

100 mL of $Na_3PO_4$ solution contains 3.45 g of sodium. The molarity of the solution is _____ $\times 10^{-2}$ mol $L^{-1}$. (Nearest integer)

[Atomic Masses - Na : 23.0 u, O : 16.0 u, P : 31.0 u]

Open solution

**Q18.** Numerical

The overall stability constant of the complex ion $[Cu(NH_3)_4]^{2+}$ is 2.1 $\times 10^{13}$. The overall dissociations constant is y $\times 10^{-14}$. Then y is _____. (Nearest integer)

Open solution

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**Q19.** Numerical

83 g of ethylene glycol dissolved in 625 g of water. The freezing point of the solution is _____ K. (Nearest integer)

Open solution

[Use : Molal Freezing point depression constant of water = 1.86 K kg $mol^{-1}$

Freezing Point of water = 273 K

Atomic masses : C : 12.0 u, O : 16.0 u, H : 1.0 u]

**Q20.** Numerical

The reaction rate for the reaction

Open solution

$[PtCl_4]^{2-} + H_2O \rightleftharpoons [Pt(H_2O)Cl_3]^{-} + Cl^{-}$

was measured as a function of concentrations of different species. It was observed that

$$\frac{-d\left[[PtCl_4]^{2-}\right]}{dt} = 4.8 \times 10^{-5}\left[[PtCl_4]^{2-}\right] - 2.4 \times 10^{-3}\left[[Pt(H_2O)Cl_3]^{-}\right][Cl^{-}].$$

where square brackets are used to denote molar concentrations. The equilibrium constant $K_c$ = _____ . (Nearest integer)

**Q21.** Numerical

A chloro compound "A".

Open solution

(i) forms aldehydes on ozonolysis followed by the hydrolysis.

(ii) when vaporized completely 1.53 g of A, gives 448 mL of vapour at STP.

The number of carbon atoms in a molecule of compound A is _____.

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## Mathematics 23 questions

**Q22.** MCQ (Single Correct Answer)

Let [t] denote the greatest integer less than or equal to t. Let
f(x) = x − [x], g(x) = 1 − x + [x], and h(x) = min{f(x), g(x)}, x ∈ [−2, 2]. Then h is :

(A) continuous in [−2, 2] but not differentiable at more than four points in (−2, 2)

(B) not continuous at exactly three points in [−2, 2]

(C) continuous in [−2, 2] but not differentiable at exactly three points in (−2, 2)

(D) not continuous at exactly four points in [−2, 2]

Open solution

**Q23.** MCQ (Single Correct Answer)

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $A^{2025} - A^{2020}$ is equal to :

(A) $A^6 - A$ (B) $A^5$ (C) $A^5 - A$ (D) $A^6$

Open solution

**Q24.** MCQ (Single Correct Answer)

The local maximum value of the function $f(x) = \left(\frac{2}{x}\right)^{x^2}$, x > 0, is

(A) $\left(2\sqrt{e}\right)^{\frac{1}{e}}$ (B) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$

(C) $(e)^{\frac{2}{e}}$ (D) 1

Open solution

**Q25.** MCQ (Single Correct Answer)

If the value of the integral
$\int_0^5 \frac{x+[x]}{e^{x-[x]}}dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in R$, $5\alpha + 6\beta = 0$, and [x] denotes the greatest integer less than or equal to x; then the value of $(\alpha + \beta)^2$ is equal to :

(A) 100 (B) 25 (C) 16 (D) 36

Open solution

**Q26.** MCQ (Single Correct Answer)

Let y(x) be the solution of the differential equation

$2x^2\,dy + (e^y - 2x)dx = 0$, x > 0. If y(e) = 1, then y(1) is equal to :

(A) 0 (B) 2 (C) $\log_e 2$ (D) $\log_e (2e)$

Open solution

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**Q27.** MCQ (Single Correct Answer)

The domain of the function $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is :

(A) $\left(-1, -\frac{1}{2}\right] \cup (0, \infty)$

(B) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$

(C) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

(D) $\left[-\frac{1}{2}, \infty\right) - \{0\}$

Open solution

**Q28.** MCQ (Single Correct Answer)

A fair die is tossed until six is obtained on it. Let x be the number of required tosses, then the conditional probability P(x $\geq$ 5 | x > 2) is :

(A) $\frac{125}{216}$

(B) $\frac{11}{36}$

(C) $\frac{5}{6}$

(D) $\frac{25}{36}$

Open solution

**Q29.** MCQ (Single Correct Answer)

If $\sum_{r=1}^{50} \tan^{-1}\frac{1}{2r^2} = p$, then the value of tan p is :

(A) $\frac{101}{102}$

(B) $\frac{50}{51}$

(C) 100

(D) $\frac{51}{50}$

Open solution

**Q30.** MCQ (Single Correct Answer)

Two fair dice are thrown. The numbers on them are taken as $\lambda$ and $\mu$, and a system of linear equations

x + y + z = 5

x + 2y + 3z = $\mu$

x + 3y + $\lambda$z = 1

is constructed. If p is the probability that the system has a unique solution and q is the probability that the system has no solution, then :

(A) $p = \frac{1}{6}$ and $q = \frac{1}{36}$

(B) $p = \frac{5}{6}$ and $q = \frac{5}{36}$

(C) $p = \frac{5}{6}$ and $q = \frac{1}{36}$

(D) $p = \frac{1}{6}$ and $q = \frac{5}{36}$

Open solution

**Q31.** MCQ (Single Correct Answer)

The locus of the mid points of the chords of the hyperbola $x^2 - y^2 = 4$, which touch the parabola $y^2 = 8x$, is :

(A) $y^3(x - 2) = x^2$

(B) $x^3(x - 2) = y^2$

(C) $y^2(x - 2) = x^3$

(D) $x^2(x - 2) = y^3$

Open solution

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**Q32.** MCQ (Single Correct Answer)

The value of

$2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is :

Open solution

(A) $\frac{1}{4\sqrt{2}}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{8\sqrt{2}}$

**Q33.** MCQ (Single Correct Answer)

If $\left(\sqrt{3}+i\right)^{100} = 2^{99}(p+iq)$, then p and q are roots of the equation :

Open solution

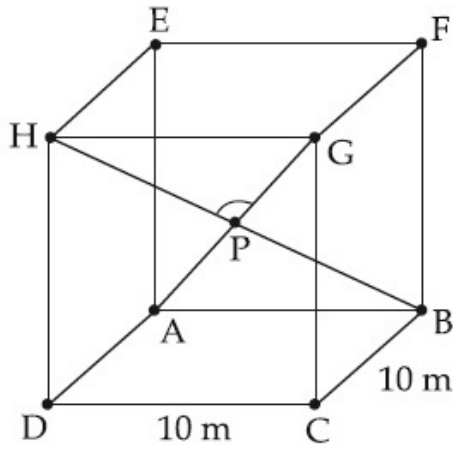
(A) $x^2-\left(\sqrt{3}-1\right)x-\sqrt{3}=0$ (B) $x^2+\left(\sqrt{3}+1\right)x+\sqrt{3}=0$

(C) $x^2+\left(\sqrt{3}-1\right)x-\sqrt{3}=0$ (D) $x^2-\left(\sqrt{3}+1\right)x+\sqrt{3}=0$

**Q34.** MCQ (Single Correct Answer)

A hall has a square floor of dimension 10 m $\times$ 10 m (see the figure) and vertical walls. If the angle GPH between the diagonals AG and BH is $\cos^{-1}\frac{1}{5}$, then the height of the hall (in meters) is :

Open solution




(A) 5 (B) $2\sqrt{10}$ (C) $5\sqrt{3}$ (D) $5\sqrt{2}$

**Q35.** MCQ (Single Correct Answer)

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)dx$ is

Open solution

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

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**Q36.** MCQ (Single Correct Answer)

A circle C touches the line x = 2y at the point (2, 1) and intersects the circle

$C_1 : x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of $C_1$. Then the diameter of C is :

(A) $7\sqrt{5}$ (B) 15 (C) $\sqrt{285}$ (D) $4\sqrt{15}$

Open solution

**Q37.** MCQ (Single Correct Answer)

$\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\frac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ is equal to :

(A) $\frac{9}{44}$ (B) $\frac{5}{24}$ (C) $\frac{1}{5}$ (D) $\frac{7}{36}$

Open solution

**Q38.** Numerical

The sum of all 3-digit numbers less than or equal to 500, that are formed without using the digit "1" and they all are multiple of 11, is _____.

Open solution

**Q39.** Numerical

Let a and b respectively be the points of local maximum and local minimum of the function f(x) = $2x^3 - 3x^2 - 12x$. If A is the total area of the region bounded by y = f(x), the x-axis and the lines x = a and x = b, then 4A is equal to _____.

Open solution

**Q40.** Numerical

If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then $\lambda$ is equal to _____.

Open solution

**Q41.** Numerical

Let $a_1, a_2, \ldots\ldots, a_{10}$ be an AP with common difference $-3$ and $b_1, b_2, \ldots\ldots\ldots, b_{10}$ be a GP with common ratio 2. Let $c_k = a_k + b_k$, k = 1, 2, ....., 10. If $c_2 = 12$ and $c_3 = 13$, then $\sum\limits_{k=1}^{10} c_k$ is equal to _____.

Open solution

**Q42.** Numerical

Let $\lambda \neq 0$ be in R. If $\alpha$ and $\beta$ are the roots of the equation $x^2 - x + 2\lambda = 0$, and $\alpha$ and $\gamma$ are the roots of equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to _____.

Open solution

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**Q43.** Numerical

Let the mean and variance of four numbers 3, 7, x and y(x > y) be 5 and 10 respectively. Then the mean of four numbers 3 + 2x, 7 + 2y, x + y and x − y is _____.

Open solution

**Q44.** Numerical

The least positive integer n such that $\frac{(2i)^n}{(1-i)^{n-2}}, i = \sqrt{-1}$ is a positive integer, is _____.

Open solution

## Physics 27 questions

**Q45.** MCQ (Single Correct Answer)

The temperature of equal masses of three different liquids x, y and z are 10°C, 20°C and 30°C respectively. The temperature of mixture when x is mixed with y is 16°C and that when y is mixed with z is 26°C. The temperature of mixture when x and z are mixed will be :

(A) 28.32°C (B) 25.62°C (C) 23.84°C (D) 20.28°C

Open solution

**Q46.** MCQ (Single Correct Answer)

The de-Broglie wavelength of a particle having kinetic energy E is $\lambda$. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value?

(A) $\frac{1}{9}$E (B) $\frac{7}{9}$E (C) E (D) $\frac{16}{9}$E

Open solution

**Q47.** MCQ (Single Correct Answer)

A particle of mass m is suspended from a ceiling through a string of length L. The particle moves in a horizontal circle of radius r such that $r = \frac{L}{\sqrt{2}}$. The speed of particle will be :

(A) $\sqrt{rg}$ (B) $\sqrt{2rg}$ (C) $2\sqrt{rg}$ (D) $\sqrt{\frac{rg}{2}}$

Open solution

**Q48.** MCQ (Single Correct Answer)

A cylindrical container of volume $4.0 \times 10^{-3}$ $m^3$ contains one mole of hydrogen and two moles of carbon dioxide. Assume the temperature of the mixture is 400 K. The pressure of the mixture of gases is :

[Take gas constant as 8.3 J $mol^{-1}$ $K^{-1}$]

(A) $249 \times 10^1$ Pa (B) $24.9 \times 10^3$ Pa (C) $24.9 \times 10^5$ Pa (D) 24.9 Pa

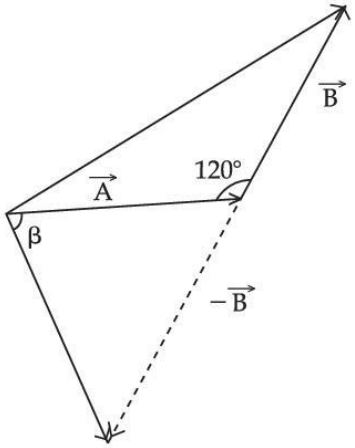
Open solution

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**Q49.** MCQ (Single Correct Answer)

The angle between vector $\left(\vec{A}\right)$ and $\left(\vec{A}-\vec{B}\right)$ is :

Open solution




(A) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$

(B) $\tan^{-1}\left(\frac{A}{0.7B}\right)$

(C) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A-B}\right)$

(D) $\tan^{-1}\left(\frac{B\cos\theta}{A-B\sin\theta}\right)$

**Q50.** MCQ (Single Correct Answer)

A light beam is described by $E = 800\sin\omega\left(t-\frac{x}{c}\right)$. An electron is allowed to move normal to the propagation of light beam with a speed of $3\times10^{7}$ ms$^{-1}$. What is the maximum magnetic force exerted on the electron?

Open solution

(A) $1.28\times10^{-18}$ N (B) $1.28\times10^{-21}$ N (C) $12.8\times10^{-17}$ N (D) $12.8\times10^{-18}$ N

**Q51.** MCQ (Single Correct Answer)

The two thin coaxial rings, each of radius 'a' and having charges +Q and −Q respectively are separated by a distance of 's'. The potential difference between the centres of the two rings is :

Open solution

(A) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$

(B) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$

(C) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$

(D) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$

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**Q52.** MCQ (Single Correct Answer)

Open solution

If you are provided a set of resistances 2Ω, 4Ω, 6Ω and 8Ω. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$.

(A) 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series

(B) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series

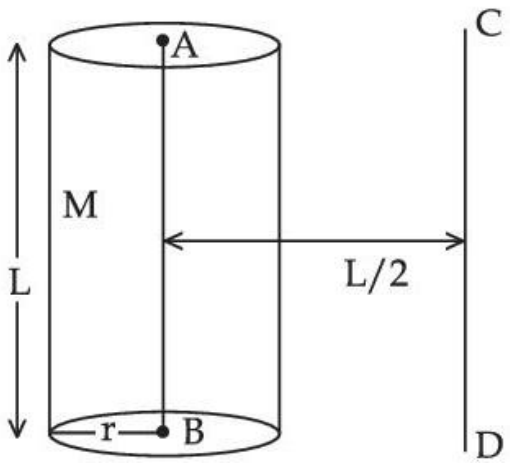
(C) 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series

(D) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series

**Q53.** MCQ (Single Correct Answer)

Open solution

The solid cylinder of length 80 cm and mass M has a radius of 20 cm. Calculate the density of the material used if the moment of inertia of the cylinder about an axis CD parallel to AB as shown in figure is 2.7 kg $m^2$.




(A) 14.9 kg/$m^3$

(B) $7.5 \times 10^1$ kg/$m^3$

(C) $7.5 \times 10^2$ kg/$m^3$

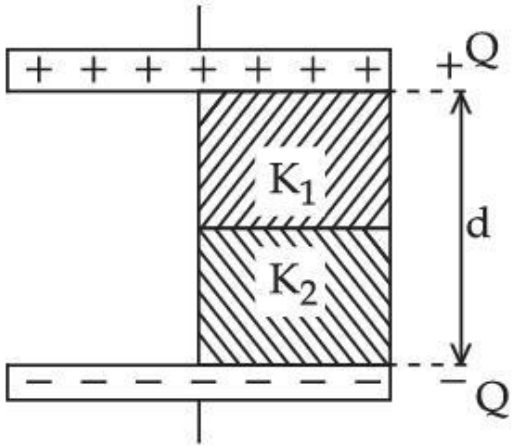
(D) $1.49 \times 10^2$ kg/$m^3$

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**Q54.** MCQ (Single Correct Answer)

A parallel plate capacitor with plate area A has separation d between the plates. Two dielectric slabs of dielectric constant $K_1$ and $K_2$ of same area A/2 and thickness d/2 are inserted in the space between the plates. The capacitance of the capacitor will be given by:

Open solution




(A) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{K_1+K_2}\right)$

(B) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{2(K_1+K_2)}\right)$

(C) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1+K_2}{K_1K_2}\right)$

(D) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{2(K_1+K_2)}{K_1K_2}\right)$

**Q55.** MCQ (Single Correct Answer)

A bomb is dropped by fighter plane flying horizontally. To an observer sitting in the plane, the trajectory of the bomb is a :

Open solution

(A) hyperbola

(B) parabola in the direction of motion of plane

(C) straight line vertically down the plane

(D) parabola in a direction opposite to the motion of plane

**Q56.** MCQ (Single Correct Answer)

An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W.

Open solution

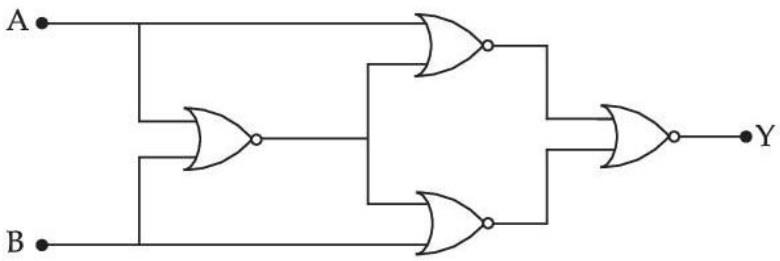
(A) 20 Ω (B) 30 Ω (C) 5 Ω (D) 10 Ω

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**Q57.** MCQ (Single Correct Answer)

Open solution

Four NOR gates are connected as shown in figure. The truth table for the given figure is :




(A)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(B)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(C)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(D)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

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**Q58.** MCQ (Single Correct Answer)

Match List - I with List - II

Open solution

| List-I | | List-II | |
|---|---|---|---|
| (a) | Magnetic Induction | (i) | $ML^2T^{-2}A^{-1}$ |
| (b) | Magnetic Flux | (ii) | $M^0L^{-1}A$ |
| (c) | Magnetic Permeability | (iii) | $MT^{-2}A^{-1}$ |
| (d) | Magnetization | (iv) | $MLT^{-2}A^{-2}$ |

Choose the most appropriate answer from the options given below :

**(A)** (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

**(B)** (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

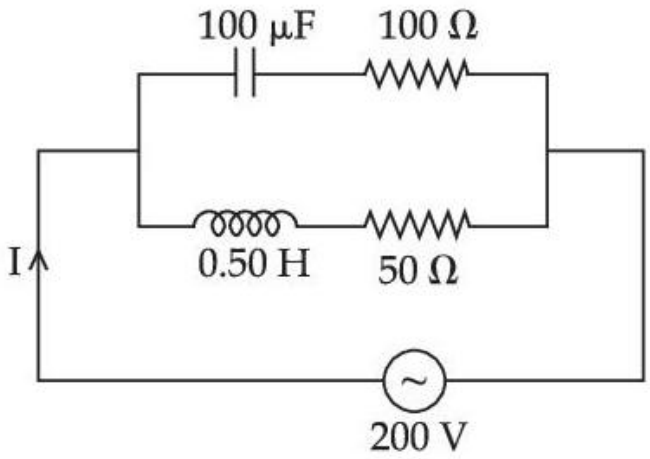
**(C)** (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

**(D)** (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

**Q59.** MCQ (Single Correct Answer)

In the given circuit the AC source has $\omega = 100$ rad $s^{-1}$. Considering the inductor and capacitor to be ideal, what will be the current I flowing through the circuit?

Open solution




**(A)** 5.9 A

**(B)** 3.16 A

**(C)** 0.94 A

**(D)** 6 A

**Q60.** MCQ (Single Correct Answer)

If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is :

Open solution

**(A)** 86.4 s

**(B)** 4.32 s

**(C)** 43.2 s

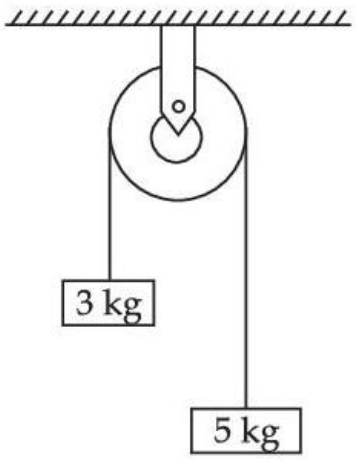
**(D)** 8.64 s

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**Q61.** MCQ (Single Correct Answer)

Open solution

Two blocks of masses 3 kg and 5 kg are connected by a metal wire going over a smooth pulley. The breaking stress of the metal is $\frac{24}{\pi} \times 10^2$ $Nm^{-2}$. What is the minimum radius of the wire ? (Take g = 10 $ms^{-2}$)




(A) 125 cm (B) 1250 cm (C) 12.5 cm (D) 1.25 cm

**Q62.** Numerical

Open solution

Two waves are simultaneously passing through a string and their equations are :

$y_1 = A_1 \sin k(x - vt)$, $y_2 = A_2 \sin k(x - vt + x_0)$. Given amplitudes $A_1$ = 12 mm and $A_2$ = 5 mm, $x_0$ = 3.5 cm and wave number k = 6.28 $cm^{-1}$. The amplitude of resulting wave will be ..... mm.

**Q63.** Numerical

Open solution

A source of light is placed in front of a screen. Intensity of light on the screen is I. Two Polaroids $P_1$ and $P_2$ are so placed in between the source of light and screen that the intensity of light on screen is I/2. $P_2$ should be rotated by an angle of (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$.

**Q64.** Numerical

Open solution

If the maximum value of accelerating potential provided by a radio frequency oscillator is 12 kV. The number of revolutions made by a proton in a cyclotron to achieve one sixth of the speed of light is .....

[$m_p = 1.67 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C, Speed of light = $3 \times 10^8$ m/s]

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**Q65.** Numerical

The acceleration due to gravity is found upto an accuracy of 4% on a planet. The energy supplied to a simple pendulum of known mass 'm' to undertake oscillations of time period T is being estimated. If time period is measured to an accuracy of 3%, the accuracy to which E is known is .....%

Open solution

**Q66.** Numerical

A circular coil of radius 8.0 cm and 20 turns is rotated about its vertical diameter with an angular speed of 50 rad $s^{-1}$ in a uniform horizontal magnetic field of $3.0 \times 10^{-2}$ T. The maximum emf induced the coil will be ..... $\times 10^{-2}$ volt (rounded off to the nearest integer)

Open solution

**Q67.** Numerical

Two simple harmonic motions are represented by the equations

$x_1 = 5\sin\left(2\pi t + \frac{\pi}{4}\right)$ and $x_2 = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$. The amplitude of second motion is ..... times the amplitude in first motion.

Open solution

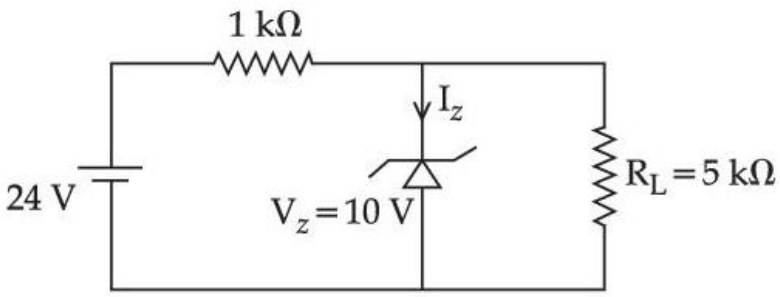
**Q68.** Numerical

A coil in the shape of an equilateral triangle of side 10 cm lies in a vertical plane between the pole pieces of permanent magnet producing a horizontal magnetic field 20 mT. The torque acting on the coil when a current of 0.2 A is passed through it and its plane becomes parallel to the magnetic field will be $\sqrt{x} \times 10^{-5}$ Nm. The value of x is .....

Open solution

**Q69.** Numerical

For the given circuit, the power across Zener diode is ..... mW.



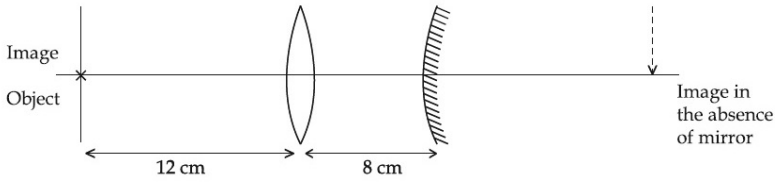

Open solution

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**Q70.** Numerical

An object is placed at a distance of 12 cm from a convex lens. A convex mirror of focal length 15 cm is placed on other side of lens at 8 cm as shown in the figure. Image of object coincides with the object.

Open solution

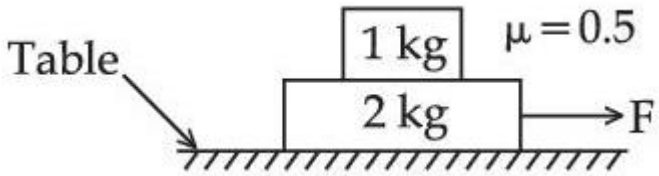



When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be ..... (cm)

**Q71.** Numerical

The coefficient of static friction between two blocks is 0.5 and the table is smooth. The maximum horizontal force that can be applied to move the blocks together is ..... N. (take g = 10 $ms^{-2}$)

Open solution




## Out of syllabus 19 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

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**Q72.** MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Photochemical smog causes cracking of rubber.

Reason (R) : Presence of ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate in photochemical smog makes it oxidizing.

Choose the most appropriate answer from the options given below :

**(A)** Both (A) and (R) are true but (R) is not the true explanation of (A)

**(B)** (A) is false but (R) is true.

**(C)** (A) is true but (R) is false.

**(D)** Both (A) and (R) are true and (R) is the true explanation of (A)

**Q73.** MCQ (Single Correct Answer) · **Chemistry**

Open solution

The interaction energy of London forces between two particles is proportional to $r^x$, where r is the distance between the particles. The value of x is :

**(A)** 3 **(B)** $-3$ **(C)** $-6$ **(D)** 6

**Q74.** MCQ (Single Correct Answer) · **Chemistry**

Open solution

The number of non-ionisable hydrogen atoms present in the final product obtained from the hydrolysis of $PCl_5$ is :

**(A)** 0 **(B)** 2 **(C)** 1 **(D)** 3

**Q75.** MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Barium carbonate is insoluble in water and is highly stable.

Reason (R) : The thermal stability of the carbonates increases with increasing cationic size.

**(A)** Both (A) and (R) are true and (R) is the true explanation of (A)

**(B)** (A) is true but (R) is false

**(C)** Both (A) and (R) are true but (R) is not the true explanation of (A)

**(D)** (A) is false but (R) is true.

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**Q76.** MCQ (Single Correct Answer) · **Chemistry**

The sol given below with negatively charged colloidal particles is :

**(A)** $FeCl_3$ added to hot water

**(B)** KI added to $AgNO_3$ solution

**(C)** $AgNO_3$ added to KI solution

**(D)** $Al_2O_3.xH_2O$ in water

Open solution

**Q77.** MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : Sphalerite is a sulphide ore of zinc and copper glance is a sulphide ore of copper.

Statement II : It is possible to separate two sulphide ores by adjusting proportion of oil to water or by using 'depressants' in a froth flotation method.

Choose the most appropriate answer from the options given below :

**(A)** Statement I is true but Statement II is false.

**(B)** Both Statement I and Statement II are true.

**(C)** Statement I is false but Statement II is true.

**(D)** Both Statement I and Statement II are false.

Open solution

**Q78.** MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Heavy water is used for the study of reaction mechanism.

Reason (R) : The rate of reaction for the cleavage of O - H bond is slower than that of O - D bond.

Choose the most appropriate answer from the options given below :

**(A)** Both (A) and (R) are true but (R) is not the true explanation of (A).

**(B)** Both (A) and (R) are true and (R) is the true explanation of (A).

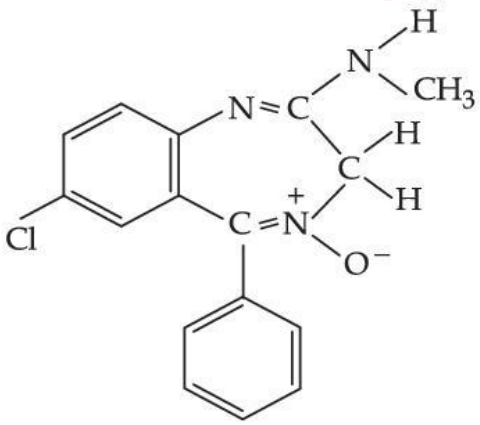
**(C)** (A) is false but (R) is true.

**(D)** (A) is true but (R) is false.

Open solution

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**Q79.** MCQ (Single Correct Answer) · **Chemistry**




Chlordiazepoxide

Open solution

The class of drug to which chlordiazepoxide with above structure belongs is :

(A) Antacid (B) Analgesic (C) Tranquilizer (D) Antibiotic

**Q80.** MCQ (Single Correct Answer) · **Chemistry**

Chalcogen group elements are :

(A) Se, Tb and Pu. (B) Se, Te and Po. (C) S, Te and Pm. (D) O, Ti and Po.

Open solution

**Q81.** MCQ (Single Correct Answer) · **Mathematics**

The point $P\left(-2\sqrt{6}, \sqrt{3}\right)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having eccentricity $\frac{\sqrt{5}}{2}$. If the tangent and normal at P to the hyperbola intersect its conjugate axis at the point Q and R respectively, then QR is equal to :

(A) $4\sqrt{3}$ (B) 6 (C) $6\sqrt{3}$ (D) $3\sqrt{6}$

Open solution

**Q82.** MCQ (Single Correct Answer) · **Mathematics**

Consider the two statements :

(S1) : $(p \rightarrow q) \vee (\sim q \rightarrow p)$ is a tautology .

(S2) : $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a fallacy.

Then :

(A) only (S1) is true. (B) both (S1) and (S2) are false.

(C) both (S1) and (S2) are true. (D) only (S2) is true.

Open solution

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**Q83.** MCQ (Single Correct Answer) · **Mathematics**

Let P be the plane passing through the point (1, 2, 3) and the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=16$ and $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$. Then which of the following points does NOT lie on P?

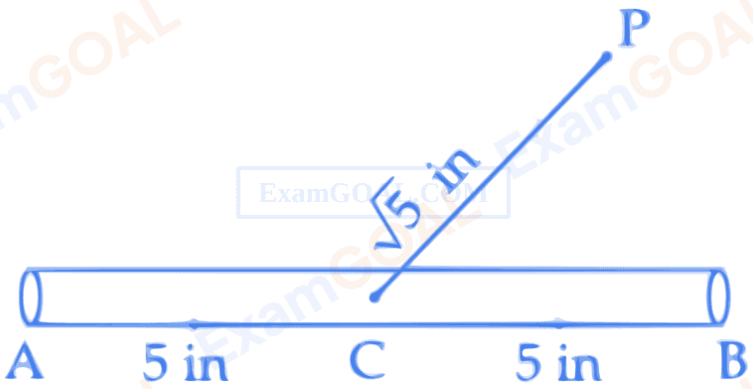
Open solution

(A) (3, 3, 2) (B) (6, −6, 2) (C) (4, 2, 2) (D) (−8, 8, 6)

**Q84.** MCQ (Single Correct Answer) · **Mathematics**

A 10 inches long pencil AB with mid point C and a small eraser P are placed on the horizontal top of a table such that PC = $\sqrt{5}$ inches and $\angle$PCB = $\tan^{-1}(2)$. The acute angle through which the pencil must be rotated about C so that the perpendicular distance between eraser and pencil becomes exactly 1 inch is :

Open solution




(A) $\tan^{-1}\left(\frac{3}{4}\right)$ (B) $\tan^{-1}(1)$

(C) $\tan^{-1}\left(\frac{4}{3}\right)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$

**Q85.** Numerical · **Mathematics**

Let Q be the foot of the perpendicular from the point P(7, −2, 13) on the plane containing the lines $\frac{x+1}{6}=\frac{y-1}{7}=\frac{z-3}{8}$ and $\frac{x-1}{3}=\frac{y-2}{5}=\frac{z-3}{7}$. Then $(PQ)^2$, is equal to _____.

Open solution

**Q86.** Numerical · **Mathematics**

Let $\binom{n}{k}$ denotes ${}^nC_k$ and $\begin{bmatrix} n \\ k \end{bmatrix}=\begin{cases}\binom{n}{k}, & if\, 0\le k\le n \\ 0, & otherwise\end{cases}$

Open solution

If $A_k=\sum_{i=0}^{9}\binom{9}{i}\begin{bmatrix}12\\12-k+i\end{bmatrix}+\sum_{i=0}^{8}\binom{8}{i}\begin{bmatrix}13\\13-k+i\end{bmatrix}$ and $A_4-A_3$ = 190 p, then p is equal to :

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**Q87.** Numerical · **Mathematics**

Let A be a $3 \times 3$ real matrix. If $\det(2\text{Adj}(2\ \text{Adj}(\text{Adj}(2A)))) = 2^{41}$, then the value of $\det(A^2)$ is equal to _____.

Open solution

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**Q88.** MCQ (Single Correct Answer) · **Physics**

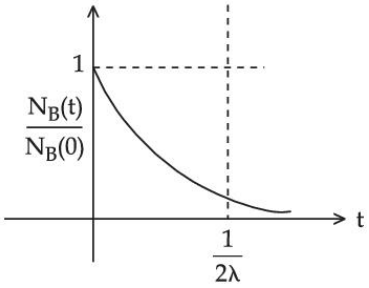
Open solution

At time t = 0, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is $\lambda$. However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t) / N_B(0)$ with respect to time t?

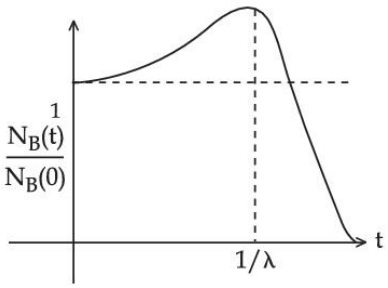
[$N_A$ (0) = No. of A atoms at t = 0

$N_B$ (0) = No. of B atoms at t = 0]

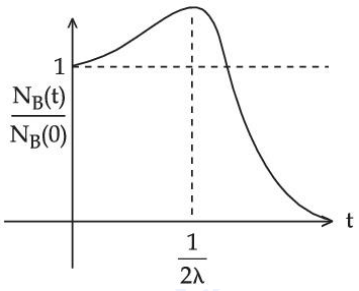
(A)



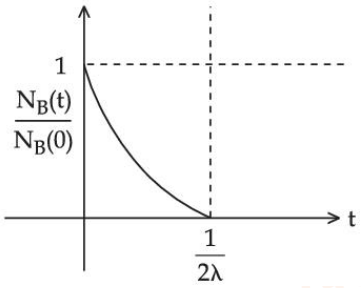

(B)




(C)




(D)




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**Q89.** MCQ (Single Correct Answer) · **Physics**

A transmitting antenna at top of a tower has a height of 50 m and the height of receiving antenna is 80 m. What is range of communication for Line of Sight (LoS) mode?

[use radius of earth = 6400 km]

Open solution

(A) 45.5 km (B) 80.2 km (C) 144.1 km (D) 57.28 km

**Q90.** MCQ (Single Correct Answer) · **Physics**

A refrigerator consumes an average 35W power to operate between temperature $-10^{\circ}C$ to $25^{\circ}C$. If there is no loss of energy then how much average heat per second does it transfer?

Open solution

(A) 263 J/s (B) 298 J/s (C) 350 J/s (D) 35 J/s

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# Answer Key

## Chemistry

| | | | | |
|---|---|---|---|---|
| 1. **B** | 2. **A** | 3. **C** | 4. **C** | 5. **D** |
| 6. **B** | 7. **B** | 8. **C** | 9. **D** | 10. **C** |
| 11. **C** | 12. **42** | 13. **182** | 14. **38** | 15. **5** |
| 16. **109** | 17. **50** | 18. **5** | 19. **269** | 20. **0** |
| 21. **3** | | | | |

## Mathematics

| | | | | |
|---|---|---|---|---|
| 22. **A** | 23. **A** | 24. **C** | 25. **B** | 26. **C** |
| 27. **D** | 28. **D** | 29. **B** | 30. **B** | 31. **C** |
| 32. **C** | 33. **A** | 34. **D** | 35. **C** | 36. **A** |
| 37. **A** | 38. **7744** | 39. **114** | 40. **5** | 41. **2021** |
| 42. **18** | 43. **12** | 44. **6** | | |

## Physics

| | | | | |
|---|---|---|---|---|
| 45. **C** | 46. **B** | 47. **A** | 48. **C** | 49. **C** |
| 50. **D** | 51. **D** | 52. **D** | 53. **D** | 54. **A** |
| 55. **C** | 56. **A** | 57. **D** | 58. **D** | 59. **B** |
| 60. **C** | 61. **C** | 62. **7** | 63. **30** | 64. **543** |
| 65. **14** | 66. **60** | 67. **2** | 68. **3** | 69. **120** |
| 70. **50** | 71. **15** | | | |

## Out of syllabus

| | | | | |
|---|---|---|---|---|
| 72. **D** | 73. **C** | 74. **A** | 75. **A** | 76. **C** |
| 77. **B** | 78. **D** | 79. **C** | 80. **B** | 81. **C** |
| 82. **C** | 83. **C** | 84. **A** | 85. **96** | 86. **49** |
| 87. **4** | 88. **C** | 89. **D** | 90. **A** | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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